FINA6229
Machine Learning in Finance

Project 1
Due Date: 12:00 PM, 19 March 2025
Stylized Facts of Stock Markets, Market Timing, and Asset Allocation

Instruction: You are encouraged to work in groups (but not across groups). However, you should independently write up and submit your own assignments through the CUHK Blackboard under the folder: *FINA6229FA>>Course Contents>>Lecture 1>>Project 1*. Demo code (as an image file) is provided under the same folder. In addition to answer the questions below, you are asked to **add some comments right after the "#" whenever it appears in the demo code**, to explain the purpose of the corresponding code.

As an example,

```
#
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)
T = monthly.shape[0]
```

you are supposed to submit the code with some comments as follows:

```
# import monthly data using the pandas module
monthly = pd.read_csv('monthly2020.csv', index_col=0, header=None)
T = monthly.shape[0]
```

1. Stylized Facts of Stock Markets

The file *monthly2022.csv* contains monthly data of portfolio returns from the combined NYSE/AMEX/Nasdaq. The first column is the date for the end of the month (in yyyymmdd format). The second column is the value-weighted market return, the third column is the equal-weighted market return. The 4th to 13th column consists of monthly returns from size decile portfolios (sorted by market capitalization, from lowest to highest).

(a) Compute the average, standard deviation, 25th, 50th, and 75th percentiles based on the monthly returns of the 12 portfolios.

(b) Compute the coefficients of skewness and (excess) kurtosis based on the monthly returns of the 12 portfolios. For each coefficient, perform a test of normality. What's your conclusion of the normality of stock returns in the U.S.?

(c) Repeat the exercises in parts (a) and (b) but based on the returns for the month of January as well as for the rest of year. What's your conclusion of stock returns in January compared with other months?

2. Market Timing and Asset Allocation

The file *PredictorData2022.xlsx* contains the data for the Goyal and Welch (RFS 2008) paper but updated to the end of 2022. The data is from Professor Amit Goyal's website. We will be using only the monthly data for this problem set.

2.1 Consider the following predictive regression:

$$r_t = \alpha + \beta x_{t-1} + \epsilon_t, \quad t = 192701, \dots, 202212,$$

where r_t is the excess (simple arithmetic) return of the value-weighted market portfolio (in excess of risk-free rate), and x_{t-1} is a predictive variable. Run this predictive regression for each one from the following set of predictive variables: *d/e*, *svar*, *dfr*, *lty*, *ltr*, *infl*, *tms*, *tbl*, *dfy*, *d/p*, *d/y*, *e/p*, *b/m*, and *ntis*. For each predictive regression, report:

(1) In-sample R^2

(2) Out-of-sample R^2 (using expanding window with an initial estimation period of 240 months)

(3) ΔCEV for a mean-variance investor with $\gamma = 3$ (and a maximum weight of 150% on the market portfolio).

Also perform this with a kitchen-sink predictive regression that includes all the predictive variables. For this multiple regression, drop *d/e* and *tms* to avoid multicollinearity issue.

2.2 Recompute the out-of-sample R^2 and ΔCEV in 2.1 by forcing the predicted market risk premium to be positive.

2.3 Consider two combination forecasts from the various predictive regressions with single-predictor. The first combination forecast is based on the median of various forecasts and the second combination forecast is based on the mean of various forecasts. Compute the out-of-sample R^2 and ΔCEV for these two combination forecasts.

3. Please submit this assignment electronically through the CUHK Blackboard as **a single Jupyter Notebook (*.ipynb) with all the outputs and answers** to the questions above and the code outputs. **Please name the submitted file with project number, your full name, student ID, and group number as**:

ProjectNumber_StudentName_StudentID_GroupNumber.ipynb,

for example: *Project1_GangLi_12345678_Group1.ipynb*.